

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

Roll No.: T013	Name: Saqlain Khan
Class: TYIT	Batch: 1
Date of Assignment: 25-07-2026	Date/Time of Submission: 25-07-2026

PRACTICAL 4

AIM :- Bayes' Rule Application

A medical test is used to detect a particular disease. The probability that a randomly selected person has the disease is 1%. If a person has the disease, the test returns positive with probability 99%. If a person does not have the disease, the test still returns positive with probability 5% (false positive rate). A person takes the test and receives a positive result. Problem Statement: Write a Python program to compute the probability that the person actually has the disease given that the test result is positive.

CODE :-

```
P_D = 0.01

P_Pos_D = 0.99

# Probability of positive test if disease does not exist

P_Pos_NotD = 0.05

# Probability of not having disease

P_NotD = 1 - P_D

# Bayes Rules

P_D_Pos = (P_Pos_D * P_D) / ((P_Pos_D * P_D) + (P_Pos_NotD * P_NotD))

print("Probability that person actually has disease : ")

print(round(P_D_Pos * 100,2), "%")

print("Saqlain Khan T013")
```

OUTPUT :-

```
Probability that person actually has disease :
16.67 %
Saqlain Khan T013
```